

Cloud Computing -- Guided Notes ANSWER KEY

Unit tp-2.11 | CompTIA Network+ N10-009 Obj. FC0-U71 2.11 | N10-008 FC0-U71 2.11

For instructor use / distribute after students complete the notes.

Question 1

Cloud computing delivers _____ resources (compute, storage, networking) over the Internet on demand. The three main service models are _____ (managed infrastructure -- VMs and storage), _____ (managed platform -- deploy apps without managing servers), and _____ (managed software -- use apps via browser).

Answer: IT; IaaS (Infrastructure as a Service); PaaS (Platform as a Service); SaaS (Software as a Service).

Question 2

The four cloud deployment models are: _____ (resources owned and operated by one organization), _____ (resources provided by a vendor to the general public over the internet), _____ (two or more organizations share cloud infrastructure), and _____ (combines private + public cloud).

Answer: Private; public; community; hybrid.

Question 3

_____ (the ability to grow resources up or out as demand increases) and _____ (the ability to shrink resources when demand decreases) are key cloud advantages. Combined, they allow organizations to pay only for what they _____ and avoid over-provisioning hardware that sits _____ during low usage periods.

Answer: Scalability; elasticity; use; idle.

Question 4

In cloud computing, the _____ model means customers pay only for what they use -- per hour, per GB, or per API call -- instead of large upfront hardware purchases. This converts capital expenditures (CapEx) to _____ expenditures (OpEx). The cloud provider owns and maintains the _____ infrastructure.

Answer: Pay-as-you-go (consumption-based); operational; physical.

Question 5

A _____ is a geographic region where a cloud provider operates multiple data centers. Within a region, there are multiple _____ (physically separate buildings) to provide _____ -- if one building loses power, the others continue. Customers choose a region to reduce _____ for their users.

Answer: Region; availability zones; redundancy (high availability); latency.

Question 6

SaaS (Software as a Service) examples include _____, _____, and _____ (list three examples). In SaaS, the provider manages the _____, _____, and _____ -- the customer just uses the

application. The customer does _____ manage any servers or platform.

Answer: Microsoft 365, Google Workspace, Salesforce (any three); infrastructure, platform, and software; not (does not).

Question 7

A CDN (Content Delivery Network) stores _____ copies of content (images, videos, web pages) at servers geographically distributed around the world called _____ Points. When a user requests content, they are served from the _____ edge node, reducing latency and load on the _____ server.

Answer: Cached; Points of Presence (PoPs) / edge nodes; nearest; origin.

Question 8 -- Real World Application

REAL WORLD: A startup is building a mobile app. They expect low traffic for the first 6 months, but if a major influencer promotes them they could see a 50x spike overnight. They have a budget of \$500/month at launch. Explain why the cloud is a better fit for this scenario than buying physical servers, and name the service model they would most likely use.

Answer: Buying physical servers requires large upfront capital expenditure (CapEx) -- a small startup spending \$500/month cannot fund servers capable of handling a 50x traffic spike. Physical servers also sit mostly idle during low-traffic months, wasting money. With IaaS or PaaS on a cloud provider (AWS, Azure, or GCP), the startup pays only for what they use during low-traffic months. When the influencer spike hits, auto-scaling can spin up additional compute instances within minutes to handle demand. After the spike, instances scale back down. This elasticity is impossible with owned physical hardware. Most app startups would use PaaS (like AWS Elastic Beanstalk or Google App Engine) or IaaS (EC2 / Compute Engine) depending on how much control they need over the environment.